

TransactiWar – Battle for Security, Compete for Supremacy.

CS6903: Network Security, 2025-26
Department of Computer Science and Engineering
Indian Institute of Technology Hyderabad

Web Application (Phase-1) Submission Deadline: March 13th, 2026, 11:59 PM

Deployment (Phase-1.1) Deadline: March 17th, 2026, 11:59 PM

War Game (Phase-2) Starts: March 18th, 2026 6:00 PM

War Game (Phase-2) Ends: March 24th, 2026 11:59 PM

(Phase-2) Report Deadline: March 26th, 2026 11:59 PM

DEADLINES ARE STRICT AND FINAL.

NO EXTENSIONS WILL BE PROVIDED IN ANY SITUATION.

LATE SUBMISSIONS ATTRACT 100% PENALTY (Phase-1)

LATE DEPLOYMENT PENALTY 50% MARKS FROM Phase-1 (Phase-1.1)

LATE SUBMISSIONS ATTRACT 20% PENALTY (Phase-2)

Phase-1: This is a group project (up to 5 members) that aims to help you understand the principles of web application security. You need to design a secure web application that implements the functionality described in each task.

Technologies allowed:

Front-end: HTML/CSS/JS/Bootstrap

Back-end: PHP + MySQL/PostgreSQL

Note: YOU CANNOT USE ANY IN-BUILD FRAMEWORKS THAT PROVIDE ANY SECURITY TO IMPLEMENT ANY OF THE TASKS.

If required, design or write your own framework. You can use cryptography libraries if needed.

Please note that we will use the Docker environment to host or evaluate your project. Therefore, ensure that your programming platform is compatible with the Docker environment and provide the necessary scripts to build the Docker image. Submissions that fail to provide such a provision will receive zero marks. **(See Deliverables 1)**

Web Application Specifications

[100 Marks]

1. User Authentication & Session Management

- a. **User Registration:**
 - i. Users register with a unique username, email, and password.
 - ii. Each new user is credited with **Rs. 100**.
- b. **Login & Logout**
- c. **Session Management**

2. Profile Management

- a. Users can update personal details (except usernames).
- b. Ability to upload and store long content (e.g., biography).
- c. Ability to upload a profile image (with secure storage).
- d. Ability to see other users' profiles.

3. User Search & Money Transfer

- a. **Search Users:** Search by username or user ID.
- b. **Money Transfer:**
 - i. Transfer money to another user by user ID.
 - ii. Prevent negative balance transactions.
 - iii. Display transaction history.
- c. **Comments**
 - i. Users can put an optional comment along with the money transfer
 - ii. The comment should be visible to the receiver

4. You must log user activity, inclusive of <Webpage, Username, Timestamp, Client's IP Address>

5. Docker Script to run your application inside the container (so automatic configuration is needed).

Deliverables - 1 (March 13th, 2026 11:59 PM):

Submission Format:

- Submit a **tar-ball/zip** file of your assignment in Google Classroom named as your **team number** (e.g., Team-01.tar.gz) containing:
 - Source Code: All PHP, HTML, CSS, JavaScript, and SQL files.
 - Script to create accounts automatically.
 - Docker Script: A Dockerfile and necessary scripts to set up and run your web application. We will use this script to host your application for the War Game.
 - Installation & Execution Guide: A README.md with instructions to run the application using Docker. Also, list the resources you referred to implement the application.

Note: Submission will only be accepted through Google Classroom. Submission via Mail, Chat or any other means will not be entertained.

All the best. Looking forward to the submissions!!... PS: Start Early.

Phase-1.1 (Deployment Phase): Students must deploy their solution on the Virtual

Machine (VM) provided by us. The deployment must be fully functional and accessible to others. The deadline specified for this phase is strict. Submissions will be evaluated based on successful deployment within the given timeframe.

Late / Failed Deployment Penalty

- **Late deployment: 50% of the marks** allocated to Phase 1 will be deducted.
- **Failure to deploy:** If the solution is not deployed on the provided VM, **100% of the phase 1** marks will be deducted.

Ensure you test your deployment in advance to avoid last-minute issues.

War Game & Security Assessment Report

[100 Marks]

Phase-2: After the submission of the web application and deployment of the project on provided VM, the War Game phase will begin, where the other teams will be attacking the opponent team's applications and will critically analyze and document the security vulnerabilities identified in their web application, the attacks conducted by other teams, the effectiveness of their security measures, and the subsequent security improvements implemented. This report will evaluate each team's understanding of web security principles, their ability to assess risks, and their ability to enhance system security post-evaluation.

You need to submit the exploited vulnerability and a screenshot of the proof in the Google form (or on another platform if ready by that time) as soon as you exploit it.

Deliverables - 2 (March 26th, 2026 11:59 PM):

Submission Format:

- Submit a **tar-ball/zip file** named as your **team number** (e.g., Team-01.tar.gz) (*only if you have used any kind of scripts*), else just upload the Report PDF without any zip (Team-01.pdf) file of your assignment in Google Classroom containing:
 - Each team must submit a comprehensive **security assessment report in the form of a PDF document** which should be named as your **team number** (e.g., Team-01.pdf)
 - A folder named "scripts" containing a subfolder with the name of the team you attacked on which would contain all the scripts that are used for attacking each opponent team's application. For Eg: scripts/team1/script.py
- *Attach Screenshots in the reports to support your claims.*
- The report should cover the following sections:
 - **Introduction**
 - Briefly describe the functionality of your web application.
 - Outline the core security mechanisms initially implemented to protect user data and transactions.
 - **War Game Attack Analysis**
 - Attacks Conducted by Other Teams on your application:
 - List the types of attacks attempted by other teams.
 - Categorize attacks based on severity and potential impact.
 - Detail whether each attack was successful or mitigated.
 - Attacks Your Team Conducted on Other's Applications:
 - Document your own team's attack strategies against peer

- applications.
 - Describe the vulnerabilities exploited and their impact.
 - Provide insights into how other teams handled security (strengths and weaknesses observed).
- **Security Patching & Enhancements**
 - Describe the security vulnerabilities identified in your application.
 - Explain the measures taken post-War Game to fix vulnerabilities.
 - Provide a before-and-after comparison of key web application security aspects.
- **Lessons Learned & Future Security Enhancements**
 - What were the biggest security flaws discovered, and what lessons were learned?
 - How would you redesign your web application to enhance security further?
 - Discuss best practices and real-world implications for secure web development.
- **Contributions:** Individual Contributions of Each Team Member.
- **References:** Include sources you have referred to. It should include references for all third-party resources used.
- **Anti-Plagiarism Statement** (Provided Below)

Note: Submission will only be accepted through Google Classroom. Submission via Mail, Chat, or any other means will not be entertained.

Penalties and Deductions:

1. If you miss any of the above instructions, including submission formats, **deductions** will be levied.
2. If other opponent team members are able to attack your team's application successfully, a penalty of **10 marks** for each security issue will be deducted for each such team. Bonus points will be awarded to teams that exploit vulnerabilities. The first three teams to exploit will receive **5, 3, and 2 marks**, respectively.
3. Deployment (Phase 1.1):
 - a. **Late deployment: 50% penalty** of the marks allocated to **Phase 1** will be deducted.
 - b. **Failure to deploy:** If the solution is not deployed on the provided VM, a **100% penalty** of the **Phase 1** marks will be deducted.
4. **Late submissions** will incur a **100% penalty** for **Phase 1** and a **20% penalty** for **Phase 2**, resulting in **zero (0)** marks where applicable.
5. **0% Plagiarism Policy (Low or High, Plagiarism is Plagiarism):**

If any plagiarism cases are found, **zero marks** and **one grade down** will be applied to the entire team.

ANTI-PLAGIARISM STATEMENT

[Include it in your PDF report]

We certify that this assignment/report is our own work, based on our personal study and/or research, and that we have acknowledged all material and sources used in its preparation, whether books, articles, packages, datasets, reports, lecture notes, or any other document, electronic or personal communication. We also certify that this assignment/report has not previously been submitted for assessment/project in any other course lab, except where specific permission has been granted from all course instructors involved, or at any other time in this course, and that we have not copied in part or whole or otherwise plagiarized the work of other students and/or persons. We pledge to uphold the principles of honesty and responsibility at CSE@IITH. In addition, we understand our responsibility to report honor violations by other students if we become aware of them.

Names <roll numbers>:

Date:

Signature: <keep your initials here>